

Automated Customer Reviews

Review Classification • Product Clustering • Generative Summarisation

An end-to-end NLP pipeline that turns thousands of raw product reviews into product recommendations.

Andre • Soledad • Benay

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Thousands of reviews, no time to read them

Product feedback is scattered across platforms and grows faster than any team can read. This project automates the reading with three tasks that turn raw reviews into structured insight:

TASK 01



Review Classify

Every review labeled Positive, Neutral or Negative with a fine-tuned transformer model.

TASK 02



Product Clustering

Thousands of products grouped into a handful of meaningful categories.

TASK 03



Generative Summarisation

AI-generated articles recommending top products and flagging common complaints.

Two very different datasets of Amazon reviews from 2023

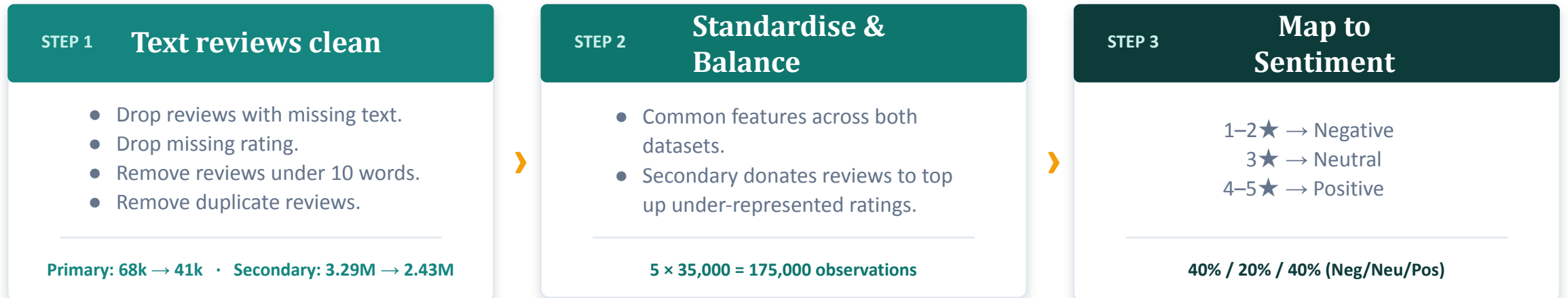
A customer's written account of their feedback about a product, usually paired with a star rating between 1 and 5.

Primary — Datafiniti				Secondary — McAuley/UCSD			
68k	92	92% rated 4-5★	19 words	3.29M	34	81% rated 4-5★	25 words
Reviews	ID Products	Rating skew	Median review length	Reviews	Product Categories	Rating skew	Median review length
Product catalog name, brand, manufacturer, categories				Product ID asin + a single generic category label			
Review content text, title, rating, date				Review content text, title, rating, timestamp			
Extra metadata purchase flag, user city/province, helpful count				Extra metadata verified purchase, helpful vote			
A rich scraped product catalog where every review is tied to a fully-described product.				A lean, review-first schema with no product name, brand, or category hierarchy.			

From raw scrapes to a balanced, model-ready dataset

After profile what each dataset contains, how it's structured, and where the two diverge, it was decide to combine them.

Three steps turn two messy raw sources into one clean, schema-aligned, class-balanced dataset ready for training.



THE MERGED DATASET

175,000 × 10

Observations × features, one row per review

Features:

asin · rating · title · text · verified_purchase · helpful_vote ·
user_id · timestamp · review_len_words · sentiment

- **Train / Test Split** — Stratified 80/20 split by sentiment: 140k train, 35k test, same balance in both.
- **Review Classification** — Clean text and balanced label feed straight into tokenisation, no further pre preparation is needed.

Fine-tuning transformers on sentiment

We fine-tuned three pretrained transformer models on the balanced 140k training set and evaluated them on the same held-out 35k test set, so no model touched the test data during training.



BASELINE

DistilBERT

A distilled, 40% smaller version of BERT, which is fast and inexpensive to fine-tune, making it a practical, low-cost starting point before testing larger models.



STRONGER ENCODER

RoBERTa

A more robustly pretrained encoder than BERT, designed to better capture nuanced and mixed sentiment in longer, less clear-cut reviews.



SELECTED ✓

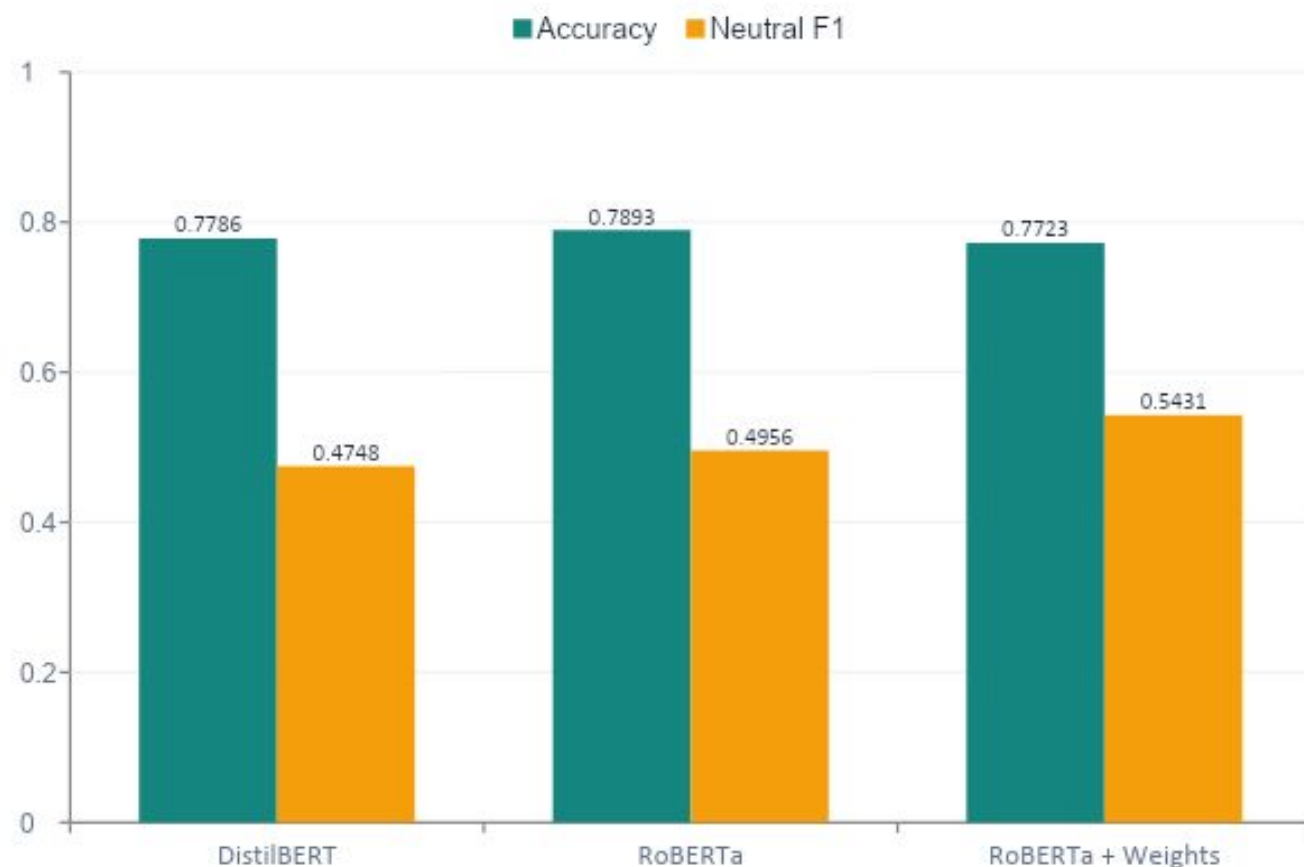
RoBERTa + Class Weights

The same encoder, but with the training loss adjusted to penalise mistakes on Neutral more heavily, which directly targets the sentiment class every model finds hardest.

Why class weights? Even on balanced data, plain training still favours the classes that are easiest to separate. Neutral reviews mix real praise and real complaint, so they resemble both Positive and Negative, so a model can reach low loss while rarely predicting Neutral at all. Weighting the loss makes Neutral mistakes costlier, pushing the model to actually learn where Neutral sits.

The accuracy–balance trade-off

After evaluated on 35,000 held-out reviews, the results All models were evaluated on the same 35,000 held-out reviews, held back entirely from training.



Selected: RoBERTa + Weights

Accuracy 77% · Neutral F1 54%

A small step down on accuracy, but Neutral recall jumps from 47% to 54%. A worthwhile trade for balance across all three classes.



The Neutral ceiling

3★ reviews genuinely mix praise and complaint, so they overlap both extremes. Most errors are Neutral leaking into Positive/Negative. So, it is a property of the data, not one model.

From raw categories to meaningful groups

Hundreds of messy category strings were reduced to a handful of intuitive meta-categories. We compared a lexical baseline against a semantic embedding approach, both with K-Means (k = 4).



TF-IDF + K-Means

Baseline

- Clusters on sparse word-frequency features
- Groups by surface words, not meaning
- Fuzzier, overlapping groups in PCA space



MiniLM Embeddings + K-Means

Selected ✓

- Sentence embeddings capture product meaning, not just shared keywords
- Groups are tight and clearly distinct from each other
- Clean visual separation, with categories named automatically via LLM

Four meta-categories



Portable Touchscreen Computing Devices

15,717 reviews

iPad · Galaxy Tab · Fire Tablet



Media Streaming Devices

5,070 reviews

Roku · Fire TV · Apple TV · Chromecast



Digital Reading Devices & Accessories

6,237 reviews

Kindle · Nook · Kobo



Home Automation Devices

7,635 reviews

Echo · Google Home · Hue · Nest · Ring

Semantically coherent clusters: streaming sticks don't leak into e-readers. Each becomes a unit the summariser writes about and the app lets users browse.

Map-reduce over thousands of reviews

Each cluster can hold 15,000+ reviews, which is far more text than a single LLM call can process. A map-reduce pipeline over DeepSeek splits the reviews into chunks, extracts findings from each, and merges them into one recommendation article per category.



Chunk

Split each cluster into shuffled
1,000 review chunks



Map

Each chunk → Prompt
structures partial findings (top
product, complaints per
product, worst product)



Reduce

Merge all partial findings →
Prompt to synthesize → One
Markdown blog article



Save

Checkpoint per cluster → Articles
are saved to JSON after each
cluster is synthesized



Every article delivers what the brief asked for: Top 3 products & their differences · top complaints per product · the worst product to avoid.

A recommendation article, auto-generated

Each article contains



Top 3 products

with key differences



Top complaints

for each product



Worst product

and why to avoid it

*~3,000 chars · grounded in real reviews ·
Markdown, ready to publish*

Portable Touchscreen Computing Devices (cluster 0)

"Fire tablets are a popular choice for families and casual users looking for an affordable, versatile device for reading, browsing and light gaming..."

Top 3 Products

Fire Tablet

Exceptional value; user-friendly interface for basic tasks.

Top complaint

Amazon Kindle Fire Tablet

Limited access to apps outside the Amazon ecosystem; Ads that cannot be removed without payment.

Product to Avoid

Fire Kids Edition Tablet Case

The Fire Kids Edition Tablet Case has received significant dissatisfaction from customers. Many express frustration over its slow performance, unresponsive touch screen, and difficulty.....

What worked, and where the challenges remain



Results

- Balanced classifier reliably distinguishes all three sentiments, with 77% accuracy and Neutral F1 at 54%.
- Embedding clusters are clean & interpretable.
- Map-reduce summaries are coherent, on-brief and grounded in reviews.
- All three tasks are delivered in one live shareable app.



Limitations

- Secondary dataset lacks product name, brand & detailed category, which limited deeper analysis to the primary set.
- Balancing was done at rating level, not sentiment level, so Neutral is still under-represented.
- Neutral reviews are intrinsically ambiguous, since collapsing mixed praise and complaint into one label erases that nuance.
- Summaries are generative and grounded in reviews, but not fact-checked against product specs.

Where we would take this next



Balance at sentiment level

Gather more genuine Neutral data, or reframe Neutral as an ordinal target.



Reduce Neutral confusion directly

Explore other models or techniques aimed at the Neutral-vs-Pos/Neg confusion itself.



Hierarchical / higher-k clustering

Surface sub-categories (accessories vs. devices) under the top-level 4 categories.



Fine-tune the generative model

Train on real review→article examples so the model learns our structure and grounding directly, instead of relying on prompting alone.

DEPLOYMENT

Everything in one interactive app

All three components are served together in a Gradio web app with a shareable public URL — usable by non-technical teams with no notebook required.



TAB 1

Sentiment Classifier

Type any text review → live Positive / Neutral / Negative prediction with per-class confidence.



TAB 2

Product Category Browser

Pick a cluster → see its products and a computed sentiment breakdown chart.



TAB 3

Recommendation Articles

Select a category → read the AI-generated summary article for it.

Thank you

From raw reviews to sentiment, categories and recommendation articles automated end to end.

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Questions welcome.